TubeLeader Micro-Tube Edition

PRECISION LASER CAM & NESTING SYSTEM (OD 0.2 MM – 10.0 MM)

Industrial CAM Solutions
For Machine Builders & OEMs

Micro-tube laser processing is fundamentally different from conventional tube fabrication. Thin wall thicknesses (< 0.05 mm), extreme thermal sensitivity, and severe rotational runout demand micron-grade trajectory resolution. **TubeLeader Micro-Tube Edition** is engineered specifically for ultra-fast (femtosecond/picosecond) and fine-fiber laser systems, providing direct CAD-to-NC automation for intricate medical hypotubes and micro-mechanical assemblies.

Core Algorithmic Innovations

Sub-Millimeter Geometry Kernel

Full 3D modeling and rotary unwrapping optimized for diameters down to **OD 0.2 mm** without floating-point precision loss. Automated parametric generation for spiral kerfs, puzzle joints, and interlocking micro-tabs.

Thermal & Stress Relief Toolpaths

Integrated micro-pulse sync, flying piercing, and alternating segmented cuts (Stress-Release Routing). Prevents heat accumulation, melting burrs, and axial tube distortion in ultra-dense cut zones.

Kerf Compensation (< 15 µm)

Sub-micron kerf width compensation tailored for tight-beam laser optics. Preserves strict mating tolerances for medical hypotube hinges and micro-connector latches.

Rotary Runout & Dynamic Compensation

Generates high-density, smooth micro-spline interpolation code. Minimizes dynamic machine vibration during high-speed rotation and eliminates stepped facet artifacts.

Target Applications & Processing Specifications

Industry Sector	Typical Tube Range	Key Applications & Micro-Features
Minimally Invasive Medical	OD 0.20 – 3.00 mm Wall 0.02 – 0.20 mm	Endoscope articulating snake joints, catheter hypotubes, cardiovascular stents, micro-biopsy forceps shafts.
Semiconductor & Microfluidics	OD 0.30 – 2.50 mm Wall 0.03 – 0.25 mm	Spring-loaded test probe barrels (Pogo Pins), sub-micron dispensing nozzles, auto-sampler needles.
Automotive & Sensors	OD 1.50 – 8.00 mm Wall 0.10 – 0.50 mm	Diesel common-rail injector needles, exhaust temperature (EGT) sensor sheaths, micro-actuator sleeves.
Electronics & Photonics	OD 0.50 – 6.00 mm Wall 0.05 – 0.30 mm	Periscope camera hollow drive shafts, micro-acoustic guide tubes, high-frequency coaxial RF shielding jackets.

OEM & System Integrator Partnership Advantages

Hardware Compatibility	Seamless post-processor support for Galil, CS-Lab, Weihong, Beckhoff, Googol, and custom PC-based motion controllers via standard or specialized G-code dialects.
Integration Models	Turnkey Standalone CAM package, OEM white-label branding, or modular DLL/SDK integration directly embedded into your machine control frontend.
Licensing Flexibility	Software dongle-free cloud activation, nodelocked keys, network floating seats, or per-machine royalty bundle agreements.

Request a Benchmark Evaluation: Provide your 3D tube design (STEP / IGES) or micro-pattern drawings. Our application engineering team will provide a comprehensive toolpath simulation, cut-time estimate, and customized post-processor test file within 48 hours.